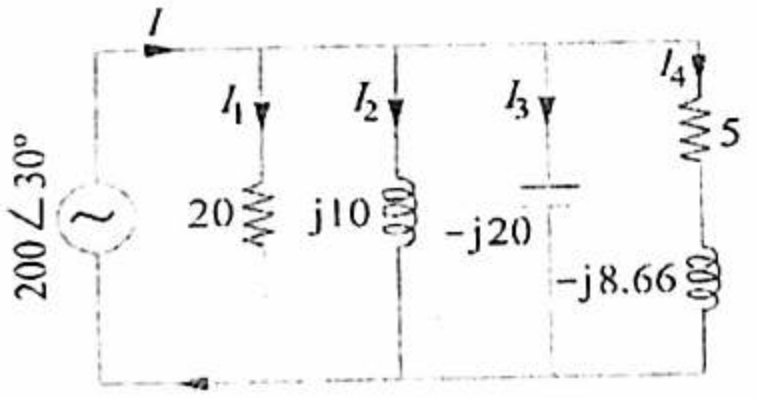
MID-TERM EXAMINATION

(Course Name: B. Tech. (Semester: I)

(October 2025) OFF LINE mode

Subject Code: BEC 101	Subject: Basics of Electrical and Electronics Engineering
Time : 1 ½Hours	Maximum Marks :30

Note: Q1 is compulsory. Attempt any two parts of Q2 and Q3

Q1	**Attempt all parts.**	**(2.5*4=10)**
	a) Define Active, Reactive and Apparent Power. Also show them on the Power Triangle.	
	b) Explain ideal voltage source and real voltage source.	
	c) Why power is not directly calculated by Superposition theorem?	
	d) A coil having an inductance of 50 mH and resistance 20 Ω is connected in series with a 35 μF capacitor across a 100 V ac supply. Calculate (1) resonance frequency of the Circuit (2) current flowing at resonance (3) value of Quality Factor.	
Q2	**Attempt any two parts.**	**(5*2=10)**
	a) Explain the concept of Resonance in series RLC circuit. Also draw the Resonance curve.	
	b) Find the value of total current and total impedance for the circuit in figure 2.a  **Figure 2.a**	
	c) A 20-Ω resistor is connected in series with an inductor, a capacitor and an ammeter across a 25-V variable frequency supply. When the frequency is 200-Hz, the current is at its maximum value of 0.5 A and the potential difference across the capacitor is 150 V. Calculate R, L and C for the circuit shown in figure 2.c	

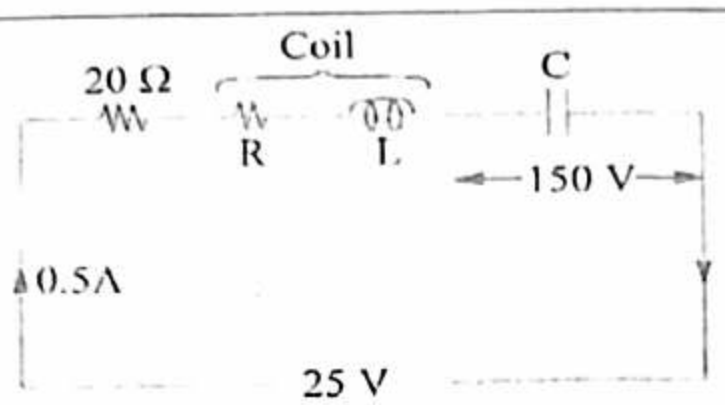


Figure 2.c

Q3	**Attempt any two parts**	**(5*2=10)**
	a) Find Thevenin equivalent circuit for the network shown in Figure. 3.a	

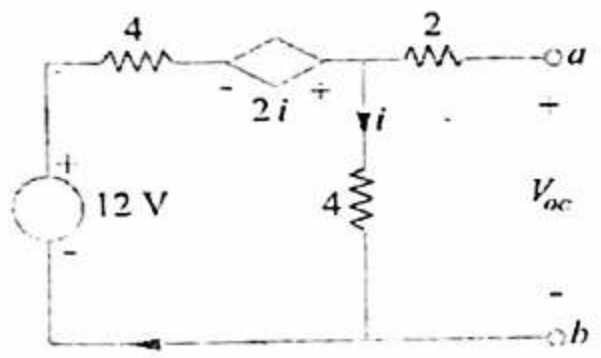


Figure. 3.a

	b) State and prove Maximum Power Theorem.	
	c) Using any method, find the value of current and voltage mention in the network as shown in figure 3.c	

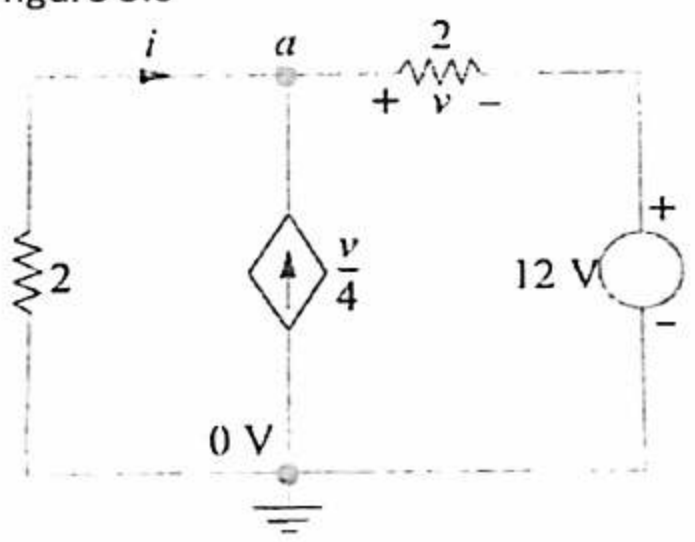


Figure 3.c